

Kiran Agarwal

Chief Substation Engineer

Power Grid Corporation of India · 13 years

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Mumbai, India

PROFESSIONAL SUMMARY

Results-driven Substation Engineer with 13+ years of experience in India's power sector. Proven track record in Protection Relay Coordination and Power Quality Analysis. Delivered complex projects across generation, transmission and distribution segments.

CORE COMPETENCIES

• Switchgear Selection	• Protection Relay Coordination	• AutoCAD Electrical
• Power Quality Analysis	• IED Configuration	• Metering & Instrumentation
• Load Flow Analysis	• Harmonic Analysis	• Demand Response Management
• PMU Installation	• Energy Trading	• Power Market Operations
• Grid Stability Analysis	• Power BI	• Big Data Analytics

PROFESSIONAL EXPERIENCE

Chief Substation Engineer2023 – Present

Power Grid Corporation of India

- Power Quality Improvement — Harmonic filter design and installation at Mumbai reducing THD from 12% to 3.5%
- Distribution Network Strengthening — R-APDRP project covering 18 feeders with automated fault detection and restoration

Substation Engineer2021 – 2023

Torrent Power

- Distribution Network Strengthening — R-APDRP project covering 16 feeders with automated fault detection and restoration
- Distribution Network Strengthening — R-APDRP project covering 19 feeders with automated fault detection and restoration
- Load Forecasting Model — ML-based short-term load forecasting achieving 9% MAPE for a 13 MW utility
- OT Cybersecurity Hardening — ICS security assessment and remediation across 2 substations per IEC 62351 standards
- Saved ₹4 Cr annually through energy audit recommendations

Protection Engineer2016 – 2021

Envision Energy India

- 400kV GIS Substation at Mumbai — Design, procurement and commissioning of 14×315 MVA auto-transformers with numerical protection
- Green Hydrogen Pilot — 13 MW electrolyzer feasibility study and FEED for renewable hydrogen production
- EV Charging Network — Design and deployment of 4 EV charging stations across Mumbai with load management integration
- 100 MW Wind Energy Project — Turbine foundation design, inter-array cabling, substation and SCADA implementation

Grid Stability Analyst2014 – 2016

Azure Power

- Renewable Integration Study — Grid impact assessment for 7 MW solar/wind addition including dynamic stability analysis
- Grid Stability Enhancement — FACTS device (SVC ±3 MVAR) installation improving voltage profile across 6 substations
- BESS Installation (3 MWh) — Grid-scale battery storage commissioning with frequency regulation and peak shaving applications

EDUCATION

M.Tech (Electrical Engineering)

2011

NIT Warangal

CERTIFICATIONS & TRAINING

- ✓ Certified Reliability Engineer (CRE)
- ✓ Python for Data Science (NPTEL)
- ✓ PMP (Project Management Professional)

References available upon request · Last updated: January 2026